MONISH D

+91 7676607637 ◊ monishpersonal609@gmail.com ◊ GitHub ◊ LinkedIn

EDUCATION

B.Tech, Computer Science Engineering (Data Science) July 2024 – Present
Vellore Institute of Technology

B.S. in Data Science and Applications August 2024 – Present
Indian Institute of Technology, Madras

EXPERIENCE

VIML Research Intern, IIT Mandi May – Jun 2026
Python, PyTorch, Hugging Face Transformers, LLaVA, Llama 2, OpenCV, Multimodal LLMs

- Built a training-light multimodal video summarization pipeline: LLaVA-1.5-7B (vision-language model) captions each frame and Llama-2-13B (LLM) scores frame importance over a 7-caption sliding window.
- Trained a lightweight transformer-style self-attention aggregator on top of the frozen 7B and 13B models; evaluated with 5-fold cross-validation on SumMe (25 videos) and TVSum (50 videos).
- Reached Kendall's τ of 0.253 and Spearman's ρ of 0.282 on held-out folds, in line with the results reported for LLMVS (CVPR 2025), while keeping the 7B and 13B models frozen.

PROJECTS

Continuum – Decision Archaeology Agent Link
Python, LangChain, LangGraph, MCP, FastAPI, TypeScript, Supabase, pgvector, Docker

- Built an LLM agent that recovers the "why" behind engineering decisions from PR threads, issues, and commits, returning source-linked answers labeled Confirmed / Inferred / Unknown; indexed 75 records from facebook/react.
- Architected a FastAPI backend with hybrid retrieval and an LLM extraction pass that structures review threads into decisions, rationale, and rejected alternatives.
- Hybrid retrieval reached 95.8% exact-symbol and 93.9% conceptual recall, versus 48.2% and 39.0% for pure vector and BM25, at 11.2 ms mean latency.

Retail Demand Intelligence Repo
R, tidyverse, ggplot2, forecast, XGBoost, Shiny, SQL

- Built an end-to-end time-series forecasting pipeline on 99K Olist orders across 9 relational tables (Sep 2016 – Oct 2018) as a reproducible R package (targets, renv), reaching 15.5% MAPE on weekly demand, about 55% better than a naive baseline.
- Built a repeat-purchase model for a marketplace where 96.9% of customers buy only once; tuning the decision threshold took it from catching 1 of 597 repeat buyers to about 12x lift over random targeting, with every modeling change validated on held-out data.

TECHNICAL SKILLS

Languages: Python, R, SQL, JavaScript, TypeScript, C++, Java
GenAI & Agentic AI: LLMs, RAG, LLM Agents, Vision-Language Models, MCP, LangChain, LangGraph, Prompt Engineering
ML & Data Science: PyTorch, TensorFlow, scikit-learn, Time-Series Forecasting, Model Evaluation, XGBoost, OpenCV
Backend & Frontend: FastAPI, REST APIs, React, Node.js, Supabase (PostgreSQL), MySQL
Tools & Deployment: Docker, Git, GitHub, Linux
Foundations: Data Structures and Algorithms, Object-Oriented Programming, System Design

CERTIFICATIONS

Designing Scalable RAG Pipelines, Scaler Masterclass — Sep 2026
Deep Learning Course, Coursera (neural networks with TensorFlow) — Jan 2026

LEADERSHIP & ACTIVITIES

Core Management Member, Android Club — 2025 – Present

- Organized Techno VIT: coordinated the event schedule, team assignments, and day-of execution across members, volunteers, and faculty coordinators.

Core Operations Member, VITSION — 2025 – Present

- Ran logistics, volunteer coordination, and on-ground operations for VITSION’s flagship events at Vibrance.